

ABISHEK MATHEWS

+1-214-682-9266 | abishekmathews7@gmail.com | in/abishekmathews | GitHub | Portfolio
U.S. Citizen (Naturalized)

EDUCATION

The University of Texas at Dallas EPPS <i>Bachelor of Science in Economics</i> GPA: 3.544	Aug. 2021 – May 2024 Richardson, TX
The University of Texas at Dallas EPPS <i>Master of Science in Social Data Analytics and Research</i>	Aug. 2024 – June 2026 Richardson, TX
National Chung Hsing University <i>Master of Science in Data Science – Graduate Institute of Data Science and Information Computing</i>	Aug. 2024 – June 2026 Taichung, Taiwan

TECHNICAL SKILLS

Languages: Python, R, Stata, MATLAB, SQL
Libraries/Frameworks: Pandas, NumPy, Scikit-learn, statsmodels, Matplotlib, Seaborn, ggplot2, R Shiny, sf, gbm
Tools & Platforms: Git, GitHub, Jupyter Notebook, RStudio, PostgreSQL, L^AT_EX, shinyapps.io
Methods: Machine Learning, Statistical Modeling, Gradient Boosting, Walk-Forward Validation, Spatial Feature Engineering, Monte Carlo Simulation, Data Visualization

MASTER'S THESIS

Predicting Residential Short-Term Rental Prices in Dallas, TX | R, Python, Gradient Boosting 2026
– Modeled localized violent-crime effects on short-term rental prices using a log-price gradient boosting model with spatial feature engineering.
– Performed neighborhood-level crime aggregation, geospatial joins, and hyperparameter tuning, and produced feature-importance analysis linking model outputs to pricing and public-safety patterns.

EXPERIENCE

Math Tutor August 2024 – Present
Mithani Tutions
– Delivered personalized one-on-one and group tutoring sessions with custom lesson plans for math readiness.

General Clerk I September 2022 – January 2023
Information Technology Coalition
– Retrieved, scanned, and verified documents for quality control and time-sensitive data accuracy.

Marketing Intern June 2022 – August 2022
Rose Gold Marketing Inc.
– Supported client account management, documentation, candidate screening, and workshop facilitation.

Math Instructor June 2021 – June 2022
Mathnasium
– Taught high school mathematics in person and remotely, supporting homework, test prep, and exam readiness.

PROJECTS

Honest Equity Momentum Backtest | Python, pandas, scikit-learn, statsmodels 2026
– Engineered a survivorship-bias-free cross-sectional equity momentum backtest: reconstructed point-in-time S&P 500 membership and built a no-lookahead execution engine validated by 47 automated tests, including explicit data-leakage guards.
– Quantified strategy decay across layered robustness checks—purged/embargoed walk-forward validation, transaction-cost modeling, Fama-French + momentum factor regressions, and deflated-Sharpe multiple-testing correction.
– Extended to a gradient-boosted model with SHAP attribution over momentum, volatility, value, and liquidity features, showing apparent gains were explained by factor exposure rather than genuine alpha.

Dallas Housing Intelligence Platform | R, Shiny, sf, gbm, SQL 2026
– Built a portfolio dashboard from thesis artifacts with a price estimator, district map, model diagnostics, and neighborhood market summaries.
– Packaged processed data, model outputs, SQL feature logic, and deployment notes for reproducible publication.

Data Forecasting Model | Python, NumPy, Anaconda June 2025
– Built a Monte Carlo simulation to project GameStop stock trajectories and evaluate future price distributions, visualizing expected returns, downside risk, and uncertainty across random price paths.

GroceryTrack | R, Shiny, PostgreSQL, shinyapps.io, LLMs June 2025
– Developed a full-stack grocery tracking app with a PostgreSQL backend and interactive R Shiny dashboard (deployed on shinyapps.io) for pantry tracking, shopping-list support, and dashboard views.

SERVICE

President September 2024 – May 2025
Beta Chi Theta National Fraternity Incorporated UT Dallas
– Led a 35-member chapter overseeing operations, finances, and governance while serving as liaison to university administration and alumni.